

Size Estimation Techniques: LOC and function point are types of **size estimation techniques** used to determine the overall size of a software project. These metrics are often inputs for other, more comprehensive estimation models, such as the COCOMO model, which then use the size to estimate effort, cost, and duration.

Line of code (LOC) measurement

LOC is simplest way to estimate the project size among all metrics available. The measure was first proposed when programs were typed on cards with one line per card. To find the LOC at the beginning of a project, divide module into sub-module and so on, until size of each module can be predicted. The project size estimation by counting the number of source instruction is an ambiguous task because there may have multiple lines of code used for commenting, header lines of codes etc. Thus, a commonly adopted convention is to count only the lines of codes that are delivered to the customers as part of the product. The LOC estimation technique has following limitations:

- The LOC will vary according to programming style i.e. complex logic may reduce some codes by the replacement of the simple logic.
- The reuse of codes will vary the program size.
- There may not be better quality on large sized programs.
- The accurate LOC are only computed after project completion.
- On project estimation, the software product not only depends on the LOC but this may vary due to analysis, design, testing etc.

Numerical

1. For example: If Estimated total LOC on any project = 33,200 Organizational average productivity = 620 LOC/pm

Labor rate per month = \$8000

Then, Cost per LOC = $(\$8000/m)/(620\text{LOC/pm}) = \13

Total project cost = Total LOC x Cost per LOC = $33,200 \times \$13$

= \$4, 31, 600

Estimated effort = Total LOC / Average productivity = $(33,200 \text{ LOC}) / (620 \text{ LOC/person month}) = 54 \text{ persons}$

2. Question – 2018 Fall (8 marks)

You are required to develop a Hotel Management, System in which the estimated lines of codes (LOC) is calculated to be 85,000, and a review of the historical data reveals that the average productivity for this type of system is 200 LOC/pm and the labor rate is Rs.

17,500 per month. What would be the estimated project cost and the estimated effort for this software package?

Solution: Here, LOC = 85000

Average productivity = 200 LOC/pm

Labor rate = Rs 17500 per month

Effort = ?

Cost = ?

We know that, $\text{Effort} = \frac{\text{LOC}}{\text{Productivity}} = \frac{85000}{200} = 425 \text{ person-month}$

Now, Project Cost = Effort * Labor rate = $425 * 17500 = \text{Rs. } 74,37,500$

3.

A university administration system is to be developed, and the estimated lines of code (LOC) are calculated to be 90,000. Based on historical data for educational domain projects, the average productivity is 400 LOC per person-month (PM), and the labor rate is Rs. 30,000 per month. Calculate the estimated effort and estimated project cost for the development of the university administration system.

Function based metric (FP): The function point (FP) metric can be used effectively as a means for measuring the functionality delivered by a system. Using historical data, the FP metric can be used to:

- Estimate the cost or effort required to design, code and test the software,
- Predicate the number of errors that will be encountered during testing,
- Forecast the number of components in implemented system
 $\text{Function point (FP)} = \text{Unadjusted function point (UFP)} \times \text{value Adjustment factor (VAF)}$

$$\text{FP} = \text{UFP} \times [0.65 + 0.01 \times \text{TDI}]$$

Where,

For UFP

Measurement parameter	Count	Weighting factor			
		Simple	Average	Complex	
Number of user inputs	×	3	4	6	=
Number of user outputs	×	4	5	7	=
Number of user inquiries	×	3	4	6	=
Number of files	×	7	10	15	=
Number of external interfaces	×	5	7	10	=
Count total	→				

- $\text{VAF (varies from 0.65 to 1.35)} = 0.65 + 0.01 \times \text{Total Degree of influence (TDI)}$
- $\text{TDI} = \text{Number of questions} \times \text{weight to each question} = 14 \times (0 \text{ to } 5) = \sum (\text{GSC})$
 Where GSC = General System Characteristics.

There are 14 GSC.

Fundamental Metrics

There are five essential fundamental metrics:

- 1) Size (in Loc)
- 2) Cost (in dollars)

- 3) Duration (in months)
- 4) Effort (in person-month)
- 5) Quality (number of faults detected)

For VAF

The F_i ($i = 1$ to 14) is GSC based on responses to the following questions:

1. Does the system require reliable backup and recovery?
2. Are specialized data communications required to transfer information to or from the application?
3. Are there distributed processing functions?
4. Is performance critical?
5. Will the system run in an existing, heavily utilized operational environment?
6. Does the system require online data entry?
7. Does the online data entry require the input transaction to be built over multiple screens or operations?
8. Are the ILFs updated online?
9. Are the inputs, outputs, files, or inquiries complex?
10. Is the internal processing complex?
11. Is the code designed to be reusable?
12. Are conversion and installation included in the design?
13. Is the system designed for multiple installations in different organizations?
14. Is the application designed to facilitate change and ease of use by the user?

Each of these questions is answered using a scale that ranges from 0 (not important or applicable) to 5 (absolutely essential).

For each complexity adjustment factor (F), give a rating on a scale of 0 to 5

0 - No influence

1 - Incidental

2 - Moderate

3 - Average

4 - Significant

5 - Essential

Formula

■ Effort = Person-Month (PM)

■ $Productivity = \frac{FP}{Effort}$ (FP/PM)

■ Total Cost = Effort * Labor Rate

■ $Cost\ per\ function = \frac{Cost}{Productivity}$ (\$)

■ $Cost\ Rate = \frac{Cost}{FP}$ (\$/FP)

■ Duration = months

■ Delivery Rate - project function point per work effort (FP/Hours)

■ Support Rate - work effort support per application function point (Hours/FP)

■ Defect Density - defects per application function point (Defect/FP).

Numerical:

Question 1) Compute the function point value for a project with following information domain characteristics:

No. of user input = 30

No. of user output = 20

No. of user inquiries = 8

No. of files = 7

No. of external interfaces = 6

Assume all complexity adjustment values are average and 14 algorithms have been counted.

Measurements Parameters	Count	Weighting Factor			Total
		Simple	Average	Complex	
1.Number of External Inputs(EI)	30	3	4	6	$30*4 = 120$
2. Number of External Output (EO)	20	4	5	7	$20*5=100$
3. Number of external inquiries (EQ)	8	3	4	6	$8*4=32$
4. Number of internal files (ILF)	7	7	10	15	$7*10=70$
5. Number of external interfaces (EIF)	6	5	7	10	$6*7=42$
				CountTotal = 364	

Since the complexity adjustment factors have average values, Now, Complexity adjustment value, F_i for average case = 3

So, sum of all F_i ($i = 1$ to 14) = $14 * 3 = 42$

$$\begin{aligned}
 \text{Function Point (FP)} &= \text{CountTotal} * [0.65 + 0.01 * \Sigma (F_i)] \\
 &= 364 * [0.65 + 0.01 * 42] \\
 &= 364 * [0.65 + 0.42] \\
 &= 364 * 1.07 \\
 &= 389.48 \\
 &= 389 \text{ FP}
 \end{aligned}$$

Therefore the function point value for the given project is 389 FP.

2. Compute the function point, productivity, documentation, and cost per function for the following data:

No. of user

input = 24 No.

of user output

= 46 No. of

user inquiries

= 8 No. of files

= 4

No. of external

interfaces = 2 Effort =

36.9 P-M (person

Month) Technical

document = 265 pages

User document = 122

pages

Cost = \$ 7744 per month

Various processing complexity

factors are: 4, 1, 0, 3, 3, 5, 4, 4,

3, 3, 2, 2, 4, 5

Solution:

Given, Effort = 36.9 P-M (person

Month) Technical document = 265

pages

User document

= 122 pages

Cost = \$ 7744

per month

Let us suppose that, given project is complex. Let's draw function point computing table:

Measurements Parameters	Count	Weighting Factor			Total
		Simple	Average	Complex	
1.Number of External Inputs(EI)	24	3	4	6	$24*6 = 144$
2. Number of External Output (EO)	46	4	5	7	$46*7= 322$
3. Number of external inquiries (EQ)	8	3	4	6	$8*6= 48$
4. Number of internal files (ILF)	4	7	10	15	$4*15=60$
5. Number of external interfaces (EIF)	2	5	7	10	$2*10=20$
				CountTotal = 594	

So, sum of all F_i ($i = 1$ to 14) = $4 + 1 + 0 + 3 + 3 + 5 + 4 + 4 + 3 + 3 + 2 + 2 + 4 + 5$
 $= 43$

Function Point (FP) = CountTotal * $[0.65 + 0.01 * \Sigma (F_i)]$
 $= 594 * [0.65 + 0.01 * 43]$
 $= 594 * [0.65 + 0.43]$
 $= 594 * 1.08$
 $= 641.52$
 $= 642 \text{ FP}$

Therefore the function point value for the given project is 642 FP.

b) Calculate the function point value for a project with the following information: 7

Information domain value	Optimistic	Likely	Pessimistic
Number of Input	20	24	30
Number of Output	12	15	22
Number of inquiries	16	22	28
Number of Files	4	4	5
Number of External Interfaces	2	2	3

Given that, all complexity adjustment values are average. The organizational average productivity for systems of this type is 6.5 FP/pm and labor rate of \$8000 per month. Also, calculate the total Cost required and Effort required.

Since the complexity adjustment factors have average values,

Now, Complexity adjustment value, F_i for average case = 3

So, sum of all F_i ($i = 1$ to 14) = $14 * 3 = 42$

Now, let's calculate the estimated value of software measurement values, i.e.

$$S = \frac{(a + 4m + b)}{6}$$

Where, Optimistic value good over bad (a), Most likely (m) and Pessimistic value – not sure value (b)

Measurements Parameters	Count	Weighting Factor		
		Optimistic (a)	Likely (m)	Pessimistic (b)
1. Number of External Inputs (EI), $S = \frac{(a+4m+b)}{6} = \frac{(20+4*24+30)}{6}$ $= 24.33 \approx 24$	24	20	24	30
2. Number of External Output (EO), $S = \frac{(a+4m+b)}{6} = \frac{(12+4*15+22)}{6} = 15.66 \approx 16$	16	12	15	22
3. Number of external inquiries (EQ), $S = \frac{(a+4m+b)}{6} = \frac{(16+4*22+28)}{6} = 22$	22	16	22	28
4. Number of internal files (ILF), $S = \frac{(a+4m+b)}{6} = \frac{(4+4*4+5)}{6} = 4.16 \approx 4$	4	4	4	5
5. Number of external interfaces (EIF), $S = \frac{(a+4m+b)}{6} = \frac{(2+4*2+3)}{6} = 2.16 \approx 2$	2	2	2	3

Let us suppose that, given project is average. **Lets draw function point computing table:**

Measurements Parameters	Count	Weighting Factor			Total
		Simple	Average	Complex	
1.Number of External Inputs(EI)	24	3	4	6	24*4 =96
2. Number of External Output (EO)	16	4	5	7	16*5 =80
3. Number of external inquiries (EQ)	22	3	4	6	22*4 = 88
4. Number of internal files (ILF)	4	7	10	15	4*10 = 40
5. Number of external interfaces (EIF)	2	5	7	10	2*7 = 14
				CountTotal = 318	

$$\begin{aligned}
\text{Function Point (FP)} &= \text{CountTotal} * [0.65 + 0.01 * \Sigma (F_i)] \\
&= 318 * [0.65 + 0.01 * 42] \\
&= 318 * [0.65 + 0.42] \\
&= 318 * 1.07 \\
&= 340.26 \\
&= 340 \text{ FP}
\end{aligned}$$

Therefore the function point value for the given project is 340 FP.

Given, Productivity = 6.5 FP/pm

Labor rate = \$8000 per month

We know that,

$$\text{Productivity} = \frac{FP}{Effort}$$

$$\text{Effort} = \frac{FP}{\text{Productivity}} = \frac{340}{6.5} = 52.31 \text{ PM} \approx \mathbf{52 \text{ PM}}$$

And, Total Cost = Effort * Labor rate

$$\begin{aligned}
&= 52 * 8000 \\
&= \mathbf{\$ 416000}
\end{aligned}$$

Assignment: Do all the numericals related to LOC and function point from question bank.

